

PHYSICS 253A, 253B, AND 253C

Quantum Field Theory

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ABSTRACT: A source-faithful textbook edition of the Physics 253abc lecture notes and recorded lectures. This pilot edition contains Chapter 1.

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Part I

Physics 253a: Quantum Field Theory I

1 Basic Generalities of Quantum Field Theory

1.1 What is quantum field theory?

What is quantum field theory? A rough answer is: quantum mechanics with relativistic locality. This answer is only useful once we explain what relativistic locality means.

There is also a basic ambiguity in the question. Working physicists use the name quantum field theory for two conceptually distinct objects. This course will discuss both. One is a local, or ultraviolet-complete, QFT; the other is an effective field theory. It is easier to begin with local QFT.

A local QFT is equipped with Poincaré symmetry, which consists of spacetime translations and Lorentz transformations. This kind of fundamental theory, first of all, is a quantum-mechanical system. Some folks might have the slight misconception that quantum field theory, going beyond quantum mechanics, is a generalization of quantum mechanics: no, quantum field theory is not a generalization of quantum mechanics. Quantum field theory is a specialization of quantum mechanics. Such a theory may have infinitely many degrees of freedom.

One requirement beyond undergraduate quantum mechanics is Poincaré symmetry. As in any quantum-mechanical system, a symmetry is represented by a unitary operator on the Hilbert space, and operators transform by conjugation. We will put these statements in the form of precise equations below.

A local QFT must also contain local operators, often called field operators. These field operators obey Poincaré covariance and microcausality. Basically, this is a precise quantum-mechanical statement that captures the idea that signals cannot propagate faster than the speed of light in a relativistic theory.

What, then, is an effective field theory? A detailed definition requires technical tools that we have not yet developed. A few general facts will suffice here.

An effective field theory is typically defined perturbatively in a coupling. So, by contrast, there are plenty of examples of quantum field theories that are completely well-defined at strong coupling. These effective field theories are typically only defined as a formal perturbative series in the coupling constant, and it may not have a well-defined limit. In particular, it may or may not converge.

Its usual formulation is a path integral, namely a functional integral over a space of fields. A path integral can also define a local fundamental theory, provided that the definition makes sense nonperturbatively. By design, an effective field theory captures long-distance or low-energy observables, with no claim beyond its regime of validity.

To make locality more concrete, consider a field operator $\hat{\phi}(x)$ associated with a spacetime point x . Merely attaching a label x to an operator is not meaningful, since one may parametrize a family of operators however one likes. Two conditions give the spacetime label its meaning. First, under a Poincaré transformation that sends one spacetime point to another, the corresponding unitary operator sends $\hat{\phi}$ to the field operator at the transformed point. Second, the operators obey microcausality: field operators at spacelike-separated, and hence causally disconnected, points commute. Poincaré covariance and microcausal-

ity are the essential criteria for locality. This concept is rather alien if you only know undergraduate quantum mechanics, but it will be central to our study of QFT.

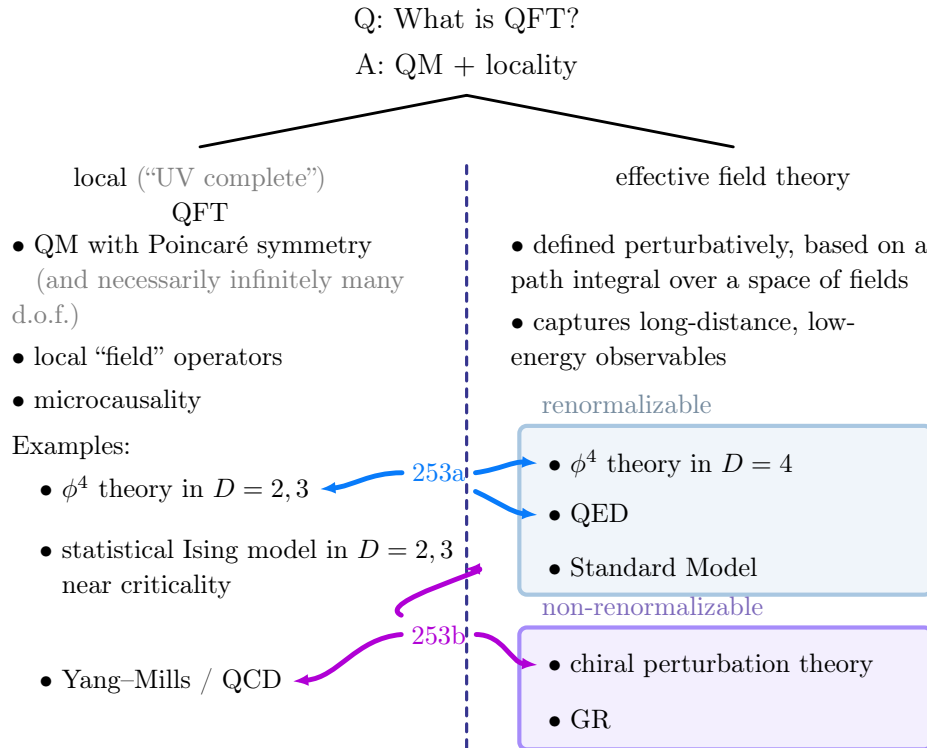


Figure 1. The opening map of local QFT and effective field theory. The colored arrows show how the topics are divided between Physics 253a and Physics 253b.

Some examples help separate local QFT from effective field theory. We will meet several of them in this course and the next. The familiar ϕ^4 theory describes a scalar field with a quartic interaction. We will introduce it in a few weeks. Let D denote the space-time dimension, including both time and space. In two or three dimensions of Minkowski spacetime, that is, in $1 + 1$ or $2 + 1$ dimensions, ϕ^4 theory is a perfectly good local QFT. In four dimensions, ϕ^4 theory is sensible only as an effective field theory.

Some of your favorite theories are unfortunately not known to be well-defined nonperturbatively as local QFTs. QED is one example. Neither is the Standard Model; both belong to effective field theory. This may make you wonder whether any local QFTs exist in four dimensions. Surprisingly, yes. Examples include Yang–Mills theory and QCD, provided there are not too many flavors. Their nonperturbative existence is far from obvious.

Local QFT also arises outside particle physics. An important example is the statistical Ising model near criticality. We will discuss this example next semester. The QFT describes observables in statistical equilibrium, where there is no time evolution, yet the statistical averages have the required local properties.

Other examples of effective field theory include quantized general relativity and chiral perturbation theory. The latter captures the low-energy interactions of pions in QCD. The

historical terms *renormalizable* and *non-renormalizable* refer to properties of the perturbative theories. We will make this distinction precise later.

1.2 Plan of Physics 253a

The course has three parts.

We begin with good old nonrelativistic quantum mechanics with a few degrees of freedom, studied in a language that may be unfamiliar from an undergraduate course but is standard in QFT. Rather than beginning with the Hamiltonian, we use the Lagrangian formulation of quantum mechanics.

In classical mechanics, the Lagrangian and Hamiltonian formulations are more or less equivalent. One may define a mechanical system by a Lagrangian, or action, and derive the Euler–Lagrange equations by extremizing the action. Alternatively, a classical state is a point in phase space, specified by coordinates and momenta. A Legendre transformation produces the Hamiltonian and its equivalent equations of motion.

When one first learns classical mechanics, these formulations may look like trivial rewritings of one another. Why bother? In quantum mechanics, the Hamiltonian is promoted from a classical function on phase space to an operator on the Hilbert space. Its eigenvalues give the energy spectrum, making it an essential object in the theory. The Lagrangian has no comparably direct operator interpretation, and it is not even clear by itself what a quantum Lagrangian should mean.

The formal Lagrangian formulation of quantum mechanics is the path integral. We will develop it together with the concept of regularization, which is standard in QFT but already present in good old quantum mechanics with only a few degrees of freedom. We then discuss perturbation theory in the path-integral language and derive Feynman diagrams. Feynman diagrams, in principle, have nothing to do with quantum field theory. They have everything to do with the perturbative expansion of a path integral. The same is true of renormalization and counterterms: they already arise in the path-integral formulation of quantum mechanics.

After developing these tools, we turn to relativistic particles and fields, beginning with scalar theories such as ϕ^4 theory. We will introduce Green functions and the asymptotic states of a scattering problem, which are more particle-physics-oriented concepts. The scattering matrix, or S-matrix, is the central observable in a large class of QFTs. The LSZ reduction formula relates it to Green functions. This relation explains why Feynman diagrams compute scattering amplitudes.

The final part of the semester concerns particles and fields with spin. We will always distinguish particles from fields, since they are very different concepts. By a field I generally mean a local field operator. Particles are particles: if it smells like a particle, looks like a particle, it will be a particle.

Both particles and fields can carry spin, but these two notions of spin have nothing to do with each other. We will explain their meanings and discuss the classification of relativistic particles. Why can there be spin-one-half fermions but no spin-one-third particles in four dimensions? We then introduce fermions and gauge bosons in preparation for QED. Standard applications include the magnetic moment of the electron and the Lamb shift.

The course plan is:

1. Lagrangian formulation of quantum mechanics:
 - path integral and regularization;
 - perturbation theory and Feynman diagrams;
 - renormalization and counter terms.
2. Relativistic particles and fields:
 - ϕ^4 theory;
 - Green functions;
 - asymptotic states;
 - S-matrix and LSZ reduction.
3. Particles and fields with spin:
 - classification of relativistic particles;
 - fermions and gauge bosons;
 - QED;
 - applications to the electron g -factor and the Lamb shift.

Canonical quantization enters before the construction of scalar field theory.

1.3 Relativistic particles without fields

We now turn to the motivation for fields.

The first question is: why do particle physicists need field theory? More precisely, why is ordinary quantum mechanics insufficient to describe particles with relativistic symmetry? The answer is not obvious.

As in any quantum-mechanical system, we have a Hilbert space whose vectors correspond to physical states. Suppose there is a vacuum state, denoted by $|\Omega\rangle$. We will assume that the vacuum is invariant under Poincaré symmetry. Otherwise a Poincaré transformation would produce a different vacuum state. In particular, the Hamiltonian generates time translations, so invariance tells us that the vacuum has zero energy:

$$H|\Omega\rangle = 0. \tag{1.1}$$

On top of the vacuum, we can have particles. The one-particle sector admits a basis labeled by spatial momentum, whose states we write as $|\mathbf{p}\rangle$. We write the Minkowski energy-momentum vector as p^μ . Its time component, the energy, is p^0 , while its spatial part is $\mathbf{p}$. Unless stated otherwise, we will work in arbitrary spacetime dimension D . A particle physicist might wonder why dimensions other than four matter. The Ising model gives one answer: it describes an actual system in solid-state physics. Thus $\mu = 0, \dots, D-1$. For the moment, one can still picture $\mathbf{p}$ as a three-dimensional vector. A relativistic particle of definite mass obeys the relativistic dispersion relation. If the vacuum energy is zero, the

Hamiltonian acting on a one-particle state of definite spatial momentum therefore gives the energy $\sqrt{\mathbf{p}^2 + m^2}$:

$$H|\mathbf{p}\rangle = \sqrt{\mathbf{p}^2 + m^2}|\mathbf{p}\rangle. \quad (1.2)$$

So far, nothing surprising. We may also consider an n -particle state whose particles have spatial momenta $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$. For now, assume these are identical, non-interacting particles of the same mass. The Hilbert space then contains states with arbitrary particle number, and the Hamiltonian acts on $|\mathbf{p}_1, \dots, \mathbf{p}_n\rangle$ with an energy equal to the sum of the individual particle energies.

$$H|\mathbf{p}_1, \dots, \mathbf{p}_n\rangle = \left(\sum_{i=1}^n \sqrt{\mathbf{p}_i^2 + m^2} \right) |\mathbf{p}_1, \dots, \mathbf{p}_n\rangle. \quad (1.3)$$

These are all the states, and we have specified how the Hamiltonian acts on each of them. We are all set. There is nothing more to do. How were these particles created? For this model, I do not care. I have declared that the states exist. Scattering will answer a different question once interactions are introduced. For the moment, this is our very simple system. Linear superpositions of these states are allowed, as always. By non-interacting, I mean that the energy of each basis state is the sum of the particle energies, with no potential contribution from mutual interactions.

The model may be rather boring, but it shows that one can construct a Lorentz-invariant quantum-mechanical system without ever talking about fields.

Creation and annihilation operators provide a convenient language for these states. We can introduce them as follows. We denote them by $a_{\mathbf{p}}^\dagger$ and $a_{\mathbf{p}}$. The spatial momentum $\mathbf{p}$ is enough to specify the particle, since its energy is fixed by the dispersion relation.

We write the creation operator as $a_{\mathbf{p}}^+ \equiv a_{\mathbf{p}}^\dagger$. The creation operator is defined by its action on states. In particular, a one-particle state is obtained by acting with $a_{\mathbf{p}}^\dagger$ on the vacuum. A two-particle state is obtained by acting with two creation operators, and so forth:

$$|\mathbf{p}\rangle = a_{\mathbf{p}}^+|\Omega\rangle, \quad |\mathbf{p}_1, \mathbf{p}_2\rangle = a_{\mathbf{p}_1}^+ a_{\mathbf{p}_2}^+|\Omega\rangle, \quad \text{etc.} \quad (1.4)$$

It remains to normalize this basis of multiparticle states. This requires the commutation relations of a and $a^\dagger$. The creation operators commute with one another, as do the annihilation operators. The mixed commutator is nonzero only when the particle momenta coincide. We choose it to be a delta function in $D - 1$ spatial dimensions. To compute $\langle \mathbf{p} | \mathbf{p}' \rangle$, write the ket as $a_{\mathbf{p}'}^\dagger |\Omega\rangle$ and the bra as $\langle \Omega | a_{\mathbf{p}}$. The commutation relation then replaces the product by the mixed commutator. This convention normalizes the overlap of two one-particle states to the same delta function:

$$[a, a] = [a^+, a^+] = 0, \quad [a_{\mathbf{p}}, a_{\mathbf{p}'}^+] = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}'), \quad \langle \mathbf{p} | \mathbf{p}' \rangle = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}'). \quad (1.5)$$

Why does this delta function sometimes appear with an extra factor? That is a matter of convention. We are free to rescale a and $a^\dagger$, which changes the normalization of the states. A one-particle momentum eigenstate is a plane wave and, strictly speaking, is not normalizable. A normalizable state is a wave packet formed by integrating these momentum

eigenstates against a function. Its overall normalization remains a convention; one could even rescale the basis by an energy-dependent factor.

The annihilation operator reduces the particle number by one, just as the creation operator increases it by one. The vacuum has zero particles by definition. Can we choose the vacuum energy however we want? Adding a constant to the Hamiltonian would shift it, but we choose the vacuum energy to be zero. This is the natural convention compatible with our Poincaré-invariant vacuum. The vacuum need not be unique. We assume that the chosen vacuum is symmetry-invariant, meaning that every generator of the continuous symmetry annihilates it. We will shortly see that H is one of these generators.

The point of introducing creation and annihilation operators is that the Hamiltonian can be written directly in terms of them, rather than defined by its action on every basis state. In terms of a and $a^\dagger$, the Hamiltonian is

$$H = \int d^{D-1}\mathbf{p} \sqrt{\mathbf{p}^2 + m^2} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}. \quad (1.6)$$

The operator $a_{\mathbf{p}}^\dagger a_{\mathbf{p}}$ measures the occupation number at momentum $\mathbf{p}$. The commutation relations verify that this Hamiltonian produces the expected energy. For massless particles, the requirement of finite particle number is much less obvious. This subtlety is tied to infrared divergences in quantum electrodynamics, and we may not reach it this semester.

We now have a well-defined Hilbert space and Hamiltonian. By design, this is a quantum-mechanical system of free relativistic particles.

If you only want to describe free particles, meaning particles that do not interact with one another, then we are done. You do not have to take this course; you can go home. We would like to describe interactions. What about interacting particles?

This might seem easy. Call the free Hamiltonian H_0 . Quantum mechanics allows us to add any Hermitian interaction term H_{int} . The interaction may contain operators such as $a^\dagger a^\dagger a$ or $aaa^\dagger$, with appropriate momentum assignments. The first can annihilate one particle and create two, thereby changing particle number.

$$H = H_0 + \underbrace{H_{\text{int}}}_{a^\dagger a^\dagger a, aaa^\dagger, \dots}. \quad (1.7)$$

Momentum kernels and Hermitian-conjugate terms are understood so that H_{int} is Hermitian. We can certainly write such terms. What is the big deal? The difficulty is preserving relativistic symmetry. It is possible to find an H_{int} that respects it, but it is not easy.

1.4 Local disturbances and causality

Consider a disturbance at one point in space at time zero. Figure 2 shows its light cone in one spatial dimension. We work in units where the speed of light is one, so light travels along rays at forty-five-degree angles. A trajectory inside the light cone is timelike and represents a signal moving slower than light. A trajectory outside the light cone would be superluminal and is forbidden in a relativistic theory. A local disturbance can produce a

response only within its light cone. This is the constraint of relativistic causality, which any relativistic theory must obey.

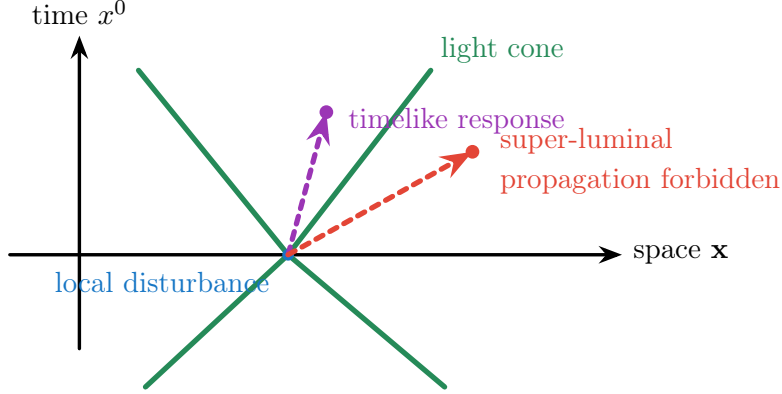


Figure 2. A local disturbance may produce a timelike response inside its light cone. A response at spacelike separation would be super-luminal.

This property is not built into the Hamiltonian formulation. Nothing there specifies a speed limit, and the influence of a generic Hamiltonian is a priori instantaneous. One can try to arrange interaction terms so that signals never propagate superluminally. It is not an easy exercise.

Local field operators give a more precise statement. Suppose an operation localized at one spacetime point changes the state of the system. In a relativistic quantum-mechanical system, such a local disturbance should be represented by a field operator.

What does field mean here? It means an operator $\hat{\phi}(x)$ that depends on a spacetime point x . Like any quantum-mechanical operator, $\hat{\phi}(x)$ acts on the Hilbert space. We may take it to be Hermitian, so that its expectation value is a physical observable. Unlike the operators familiar from undergraduate quantum mechanics, this is an entire family of operators labeled by a parameter. Here x is a spacetime point, rather than an operator. The operator is $\hat{\phi}$. Thus $x^\mu = (x^0, \mathbf{x})$ is only a label. Associated with each such point is an operator $\hat{\phi}(x)$ representing a local disturbance of the system.

1.5 Poincaré covariance

Now, this field operator should obey some properties. We said these in words earlier, but let me now say this more precisely. If I look at $\hat{\phi}(x)$ and $\hat{\phi}(x')$, at different spacetime points x and x' , these two operators are supposed to be related by a Poincaré symmetry that sends x to x' . Since we are in Minkowski spacetime, I can always find some Poincaré transformation that takes x to x' . It could be a translation, a rotation, or whatever. The choice is highly non-unique. You'd expect that this symmetry taking x to x' should also take $\hat{\phi}(x)$ to $\hat{\phi}(x')$. This is a reasonable requirement if you want to say that $\hat{\phi}(x)$ is indeed tied to that spacetime point in some sense.

Consider a transformation that takes x to x' :

$$x^\mu \rightsquigarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu. \quad (1.8)$$

This says that you can shift x by some constant spacetime vector a^μ . You can also rotate or boost, which is a Lorentz transformation $\Lambda^\mu{}_\nu$. The transformation matrix $\Lambda^\mu{}_\nu$ is supposed to preserve the Lorentzian metric, so

$$\eta^{\alpha\beta}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta = \eta^{\mu\nu}, \quad \eta^{\mu\nu} = \text{diag}(-1, 1, \dots, 1), \quad (1.9)$$

where we're using the mostly-positive signature for the Lorentzian metric in D dimensions.

This is the usual Poincaré transformation from special relativity. In elementary mechanics, Poincaré symmetry acts on spacetime points. In quantum mechanics, the same symmetry is represented by a unitary operator. Here the Poincaré symmetry acts on states in the Hilbert space. The corresponding unitary operator will be denoted by U . Since the transformation is specified by the Lorentz matrix Λ and the translation a , we write this operator as $U(\Lambda, a)$.

In particular, we need to say how U acts on a field operator. A state is transformed by U , while an operator is transformed by conjugation with the same unitary operator, as in $U\hat{\phi}(x)U^{-1}$.

$$\hat{\phi}(\Lambda x + a) = U(\Lambda, a) \hat{\phi}(x) (U(\Lambda, a))^{-1}. \quad (1.10)$$

We require this similarity-transformed operator to equal the field operator at the new point $\Lambda x + a$. Spacetime indices are implicit in this shorthand. Thus $\hat{\phi}(x)$ and $\hat{\phi}(x')$, with $x' = \Lambda x + a$, are related by the symmetry.

The displayed law assumes that $\hat{\phi}$ is a scalar field operator. We will consider fields with other transformation laws later. Whether an operator is scalar has nothing to do with which particles occur in the theory. A theory containing fermions or gauge bosons may still have scalar field operators.

What do we know about $U(\Lambda, a)$? So far, only that it depends on Λ and a . One way to study a continuous symmetry is to regard a finite transformation as a composition of infinitesimal ones. In this case it suffices to focus on the infinitesimal transformation. Take $\Lambda^\mu{}_\nu$ to be the identity plus a small variation $\omega^\mu{}_\nu$, and write the infinitesimal translation as $a^\mu = \epsilon^\mu$.

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad a^\mu = \epsilon^\mu. \quad (1.11)$$

Substituting $\Lambda = \delta + \omega$ into the metric-preserving condition and expanding to first order shows that $\omega_{\mu\nu}$, with one index lowered by $\eta_{\mu\nu}$, is antisymmetric: $\omega_{\mu\nu} = -\omega_{\nu\mu}$.

The unitary operator U is also close to the identity. To first order in ω and ϵ , it is the identity plus an anti-Hermitian operator. A convenient expansion contains one term proportional to ϵ^μ and another proportional to the antisymmetric tensor $\omega_{\mu\nu}$. The factor of one half accounts for its antisymmetry. The operators $\hat{P}$ and $\hat{J}$ are defined by this expansion. Pulling out a factor of i makes the $\hat{P}$'s and $\hat{J}$'s Hermitian. The operator $\hat{P}_\mu$ is the energy-momentum operator. The operators $\hat{J}^{\mu\nu}$ generate Lorentz boosts and rotations.

$$U(\Lambda, a) = 1 - i\epsilon^\mu \hat{P}_\mu + \frac{i}{2}\omega_{\mu\nu} \hat{J}^{\mu\nu}. \quad (1.12)$$

We will not derive here the commutators of the P 's and J 's. They follow from the composition law for Poincaré transformations. In particular, $U(\Lambda', a')$ composed with

$U(\Lambda, a)$ first sends x to $\Lambda x + a$, and then applies Λ' and a' . The resulting transformation is $U(\Lambda'\Lambda, \Lambda'a + a')$.

$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda'a + a'). \quad (1.13)$$

Expanding this law near the identity gives the commutation relations of the P 's and J 's. They are fixed by the requirement that the U 's represent Poincaré transformations.

$$\begin{aligned} [P, P] &= 0, \\ [P^\mu, J^{\rho\sigma}] &= -i(\eta^{\mu\rho}P^\sigma - (\rho \leftrightarrow \sigma)), \\ [J^{\mu\nu}, J^{\rho\sigma}] &= -i(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - (\rho \leftrightarrow \sigma)). \end{aligned} \quad (1.14)$$

Energy and momentum are the conserved charges associated with translations in time and space. We will give a systematic field-theoretic account using the Noether procedure in a few lectures. That procedure expresses the P 's and J 's as spatial integrals of the stress-energy tensor. The relation to the stress-energy tensor is discussed in Chapter 7.

1.6 Microcausality and the free scalar field

The covariance equation explains what it means for $\hat{\phi}$ to be tied to the spacetime point x : the Poincaré transformation represented by U carries $\hat{\phi}(x)$ to $\hat{\phi}(\Lambda x + a)$.

Locality adds a further requirement: microcausality. For the present scalar operator, we demand that $\hat{\phi}(x)$ and $\hat{\phi}(y)$ commute whenever the spacetime points x and y are spacelike separated. With our mostly-positive convention, the Lorentzian inner product is $(x - y)^2 = -(x^0 - y^0)^2 + (\mathbf{x} - \mathbf{y})^2$. Spacelike separation means that this quantity is positive:

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \quad \text{whenever} \quad (x - y)^2 \equiv -(x^0 - y^0)^2 + (\mathbf{x} - \mathbf{y})^2 > 0. \quad (1.15)$$

Here and below, the speed of light is set to one. This is microcausality for the local field operator $\hat{\phi}$.

Is the existence of such a local field obvious? It is a nontrivial constraint on the quantum system to require an operator $\hat{\phi}(x)$ with both covariance and microcausality.

Does this apply to our system of free particles? Intuitively it should: the particles are non-interacting and propagate with the relativistic dispersion relation. The real question is whether local field operators can be written down in this theory. Consider again this simple quantum-mechanical model of free relativistic particles.

We have already written its Hamiltonian:

$$H = \int d^{D-1} \mathbf{p} \sqrt{\mathbf{p}^2 + m^2} a_{\mathbf{p}}^+ a_{\mathbf{p}}. \quad (1.16)$$

The momentum operator is written similarly:

$$\mathbf{P} = \int d^{D-1} \mathbf{p} \mathbf{p} a_{\mathbf{p}}^+ a_{\mathbf{p}}. \quad (1.17)$$

Lower-case p is the integration variable, while capital P denotes the operator. Together the Hamiltonian and spatial momentum form the energy-momentum operator $\hat{P}^\mu = (H, \mathbf{P})$.

$$\hat{P}^\mu = (H, \mathbf{P}). \quad (1.18)$$

These are some of the symmetry generators of the model. The J 's are more complicated, and their construction will become clear later.

But the key question you might ask is: what the heck is $\hat{\phi}(x)$ in this model? Does it even exist? That is, does an operator $\hat{\phi}(x)$ exist that satisfies all of the nice properties we just demanded? The answer is yes, but it's not totally obvious. We'll have a different way to understand this later, but for now I'll just write down the answer.

It exists and is not unique. One choice can be written in terms of the $a_{\mathbf{p}}$'s. The overall normalization is conventional. The essential quantity $\omega_{\mathbf{p}}$ is the positive-energy relativistic dispersion relation for a particle of mass m :

$$\hat{\phi}(x) = \int \frac{d^{D-1}\mathbf{p}}{\sqrt{(2\pi)^{D-1}2\omega_{\mathbf{p}}}} (a_{\mathbf{p}}e^{ip \cdot x} + a_{\mathbf{p}}^+e^{-ip \cdot x}). \quad (1.19)$$

$$p \cdot x \equiv \mathbf{p} \cdot \mathbf{x} - \omega_{\mathbf{p}}x^0, \quad \omega_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + m^2}. \quad (1.20)$$

The field operator is the momentum integral above, with $p \cdot x = \mathbf{p} \cdot \mathbf{x} - \omega_{\mathbf{p}}x^0$. The integral runs over $\mathbf{p}$, and $\omega_{\mathbf{p}}$ is its energy according to the dispersion relation. We will derive this construction systematically later.

This construction has all the nice properties we demanded. It transforms correctly under Poincaré transformations and, importantly, obeys microcausality.

To check microcausality, compute the commutator:

$$[\hat{\phi}(x), \hat{\phi}(y)] = \int \frac{d^{D-1}\mathbf{p}}{(2\pi)^{D-1}2\omega_{\mathbf{p}}} \cdot [e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)}]. \quad (1.21)$$

$$[\hat{\phi}(x), \hat{\phi}(y)] = \int \underbrace{\frac{d^D p}{(2\pi)^{D-1}} \theta(p^0) \delta(p^2 + m^2)}_{\text{invariant for } \Lambda \in SO^+(1, D-1)} \underbrace{[e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)}]}_{\text{scalar under simultaneous } p \rightarrow \Lambda p, (x-y) \rightarrow \Lambda(x-y)}. \quad (1.22)$$

Under a proper orthochronous Lorentz transformation, both p and $z = x - y$ transform; changing the integration variable leaves the integral invariant. For spacelike z , choose a Lorentz frame with $z^0 = 0$ and $\mathbf{z} \neq 0$. The integrand is odd under $\mathbf{p} \rightarrow -\mathbf{p}$, so the integral vanishes.

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \quad \text{for} \quad (x - y)^2 > 0. \quad (1.23)$$

The free field is also required to obey the covariance property below. Its direct verification is deferred until the systematic construction.

$$U(\Lambda, a) \hat{\phi}(x) (U(\Lambda, a))^{-1} = \hat{\phi}(\Lambda x + a). \quad (1.24)$$

1.7 Postulates and the next formulation

We therefore postulate that a system of interacting relativistic particles has the same structure:

- a Hilbert space $\mathcal{H}$;

- Poincaré symmetry $U(\Lambda, a)$;
- a Poincaré-invariant vacuum $|\Omega\rangle$;
- local “field” operators $\hat{\phi}(x)$ obeying Poincaré covariance and microcausality.

An interacting system with the same properties is difficult to construct in the Hamiltonian formalism. The basic reason is that this formalism singles out time from space, so Poincaré symmetry is not manifest. For example, our integrals over spatial momentum do not look Lorentz covariant, even when the underlying physics is Lorentz invariant. Operators can be constructed, but checking the symmetry is not trivial. It will therefore be useful to have a formulation in which Poincaré symmetry is manifest. This is why the Lagrangian formalism will be useful. We will next formulate quantum mechanics in the Lagrangian language.

How was the displayed field operator constructed? At this stage, it should be regarded as the answer suggested by the desired properties. One can check that it works; a systematic construction comes later.

Editorial note. Problem Set 1 on source physical pages 15–19 is deferred for later integration into the exercises appendix.